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FACULTY OF CONTROL, ROBOTICS
AND ELECTRICAL ENGINEERING

AUTOMATIC CONTROL AND ROBOTICS



AUTONOMOUS CAR CONTROLLED BY THE DIRECTION OF THE SOUND ELECTRICAL DRIVES

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PROJECT OVERVIEW

The goal of this project was to design and build a small autonomous vehicle capable of detecting sound from its surroundings and navigating toward the direction of the loudest source. The core assumption was as follows: the vehicle is equipped with multiple microphones placed at different positions on its chassis. By continuously comparing the amplitude of the signals received by each microphone, the control system would determine the dominant sound direction and move the vehicle in this direction.

The vehicle was built around an ESP32-S3 microcontroller, which handles both signal acquisition from the KY-037 microphone modules and motor control via an L298N H-bridge driver. Two JGA25-370 DC motors with built-in encoders provide the drive and enable closed-loop speed control.

The concept is straightforward : sample the microphone outputs for some time (e.g. 5s), identify which side receives the louder signal, and command the motors to turn the vehicle in that direction for some time (e.g. 5s). However, as discussed in detail in the conclusions, the practical realisation of this idea proved significantly more challenging than anticipated, primarily due to the limitations of the selected microphone module.

The L298N is a dual H-bridge motor driver capable of driving two DC motors independently with up to 2 A per channel. It accepts logic-level PWM signals from the microcontroller to control both the speed and direction of each motor.

In this project, the L298N served as the power stage between the ESP32-S3 and the two JGA25-370 motors. The enable pins were connected to PWM-capable GPIO outputs on the microcontroller, and the direction pins were controlled with standard digital outputs. The module's built-in flyback diodes provide protection against the inductive spikes generated when motor windings switch. The supply voltage for the motor side was taken directly from the AA battery pack.

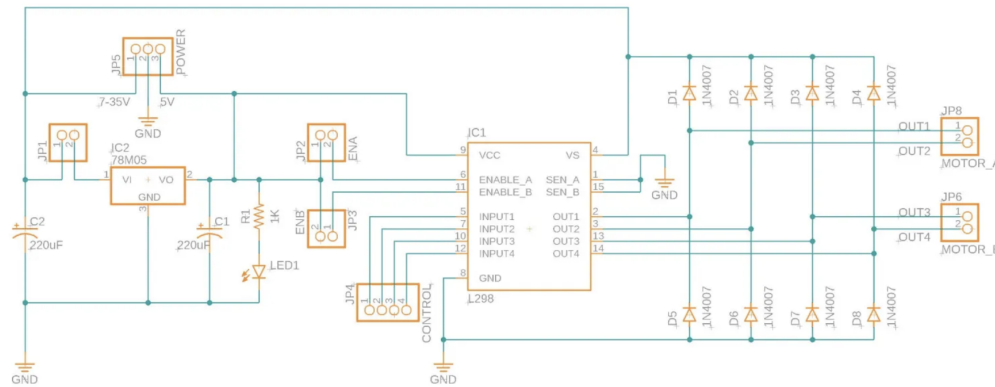


Fig. 2. Schematic of L298N Motor Driver

2.4 ESP32-S3-DEVKITC-1 (N8R8) MICROCONTROLLER

The ESP32-S3-DevKitC-1 with the N8R8 module was chosen as the main processing unit. This board is based on the Xtensa LX7 dual-core processor running at up to 240 MHz and features integrated Wi-Fi (802.11 b/g/n) and Bluetooth 5.0. The N8R8 variant includes 8 MB of PSRAM, which is useful for buffering audio data if needed.

Relevant peripherals used in this project include:

- ADC: for reading the analogue output of the KY-037 modules
- GPIO with interrupt support: for reading encoder pulses from the motors
- MCPWM or LEDC peripheral: for generating PWM signals to the L298N
- Wi-Fi: used in the final implementation for remote vehicle control via a simple web interface

2.5 POWER SUPPLY - 4× AA BATTERY PACK

The vehicle is powered by four AA batteries connected in series, providing a nominal voltage of approximately 6 V. This voltage is well-matched to the rated operating voltage of the JGA25-370 motors and falls within the acceptable input range of the L298N driver's motor supply.

The ESP32-S3 development board and the KY-037 modules are powered from the 3.3 V regulated output provided by the on-board voltage regulator of the DevKit, which in turn is supplied from the battery pack.

CIRCUIT SCHEMATIC

3.1 CONNECTION DIAGRAM

The full wiring diagram showing connections between the ESP32-S3, L298N motor driver, JGA25-370 motors (with encoders), KY-037 microphone modules, and the battery pack is provided below.

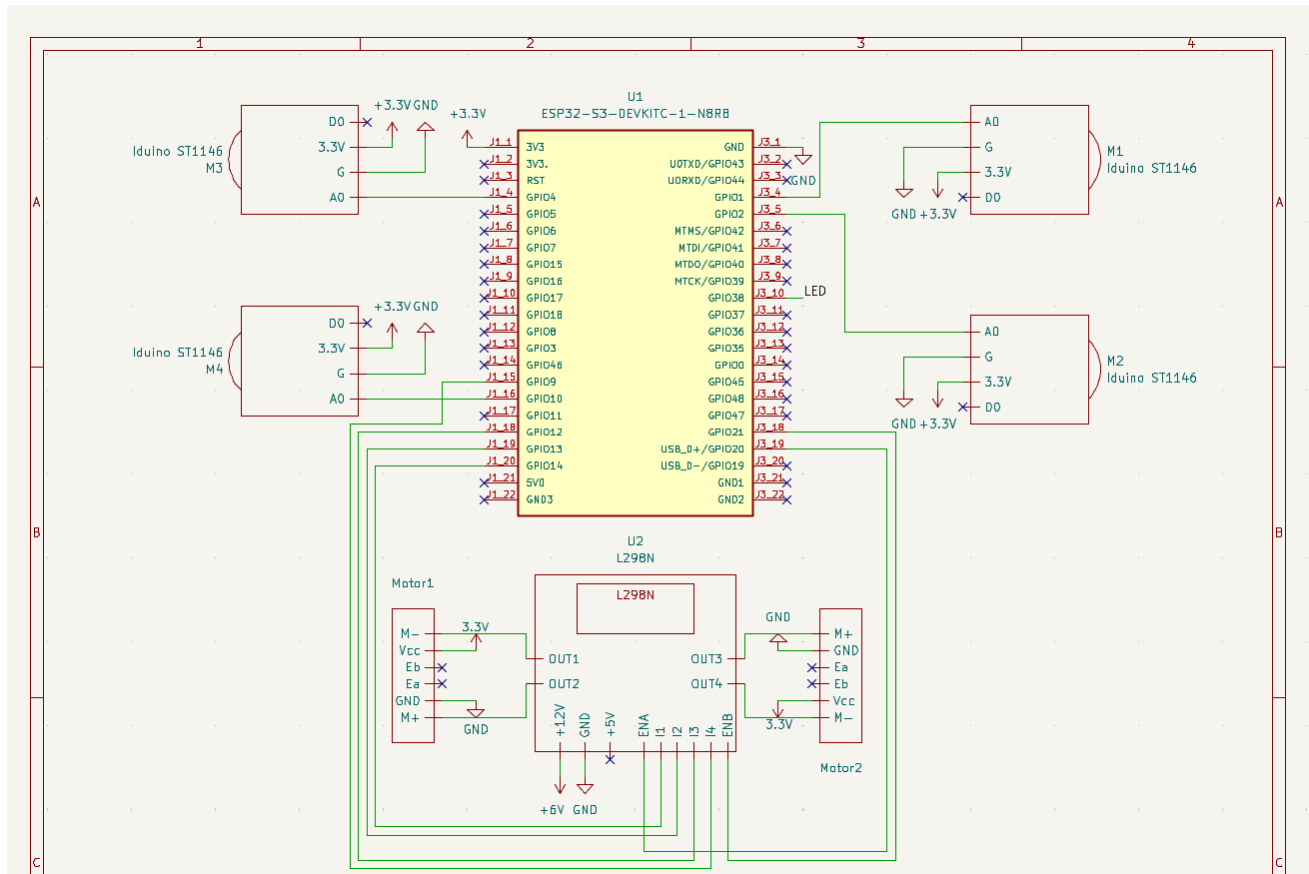


Fig. 3. KiCad schematic of whole system

MEASUREMENTS

4.1 MICROPHONE SIGNAL ANALYSIS



Fig. 4. Microphone Measurement Streamed Live on the Website with Decision Making Results

To evaluate the usability of the KY-037 module for directional sound detection, the analogue output was sampled at a fixed rate using the ESP32-S3 ADC. Measurements were taken under various conditions: background noise, a clap at a known distance, and sustained tones at different frequencies.

The plots confirm that while transient events (such as a sharp clap or blow) produce a detectable voltage change, in case of quieter sounds (such as talking) the signal is too small, not even always visible for reliable amplitude comparison between two physically separated modules.

4.2 MOTOR SPEED AND ENCODER MEASUREMENTS

The performance of the drive system was verified by recording encoder pulse counts over time and calculating the resulting wheel speed. Step responses for different PWM duty cycles were also collected to characterise the motor dynamics.

The motors responded predictably to PWM commands, and the encoder feedback provided stable speed estimates, confirming that the drive subsystem is suitable for implementing closed-loop control.

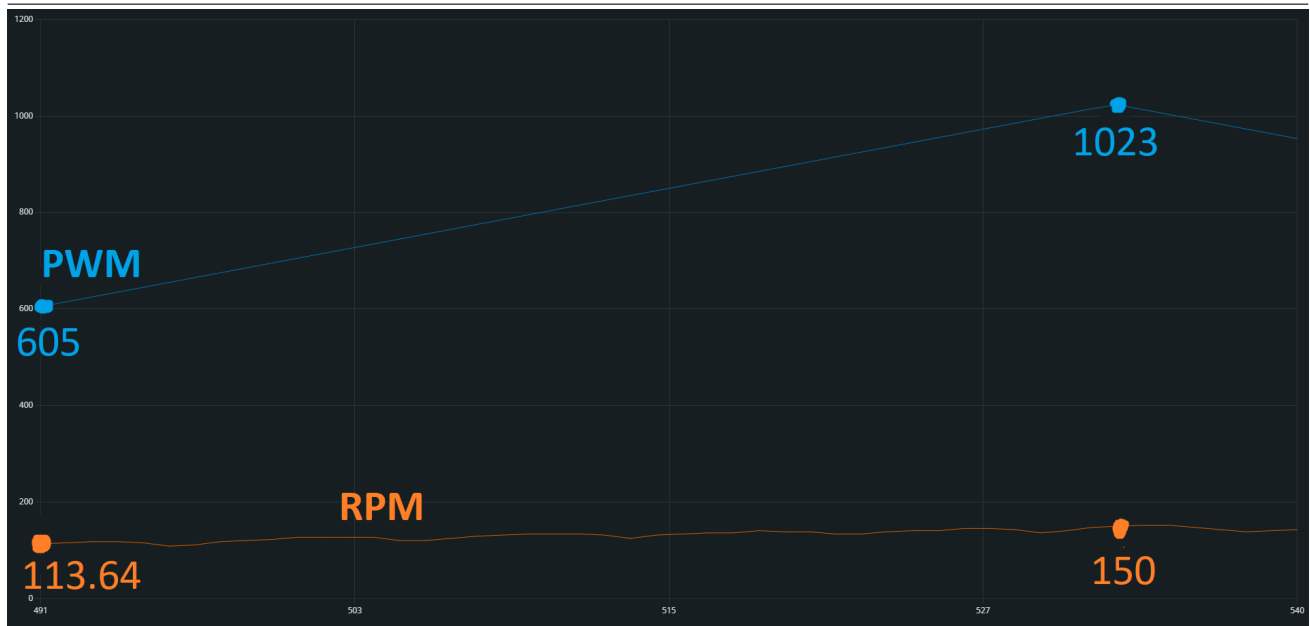


Fig. 5. Motor speed response for a change in PWM duty cycle.

FIRMWARE

5.1 CODE DESCRIPTION

The firmware was developed for the ESP32-S3 using the Arduino framework and is responsible for:

- Acquiring analog signals from four directional microphones.
- Determining the direction of the loudest sound source.
- Controlling a DC motor driver (L298N) to move the vehicle according to the detected direction.
- Driving an RGB NeoPixel LED for visual feedback.
- Hosting a Wi-Fi Access Point and web server for real-time monitoring.
- Managing the operating sequence using a finite-state machine.

The system operates in three states:

1. **COLLECTING** – microphone data are sampled for 5 seconds and peak values are stored.
2. **EXECUTING** – the vehicle performs the detected movement command.
3. **SHOWING** – used when no valid sound direction is detected; the result is displayed before a new cycle begins.

The firmware continuously compares the maximum sound amplitudes measured by the four microphones and determines the dominant direction (left, right, front, or rear). Once a valid direction is identified, the corresponding motor control command is executed automatically.

5.2 BLOCK DIAGRAM

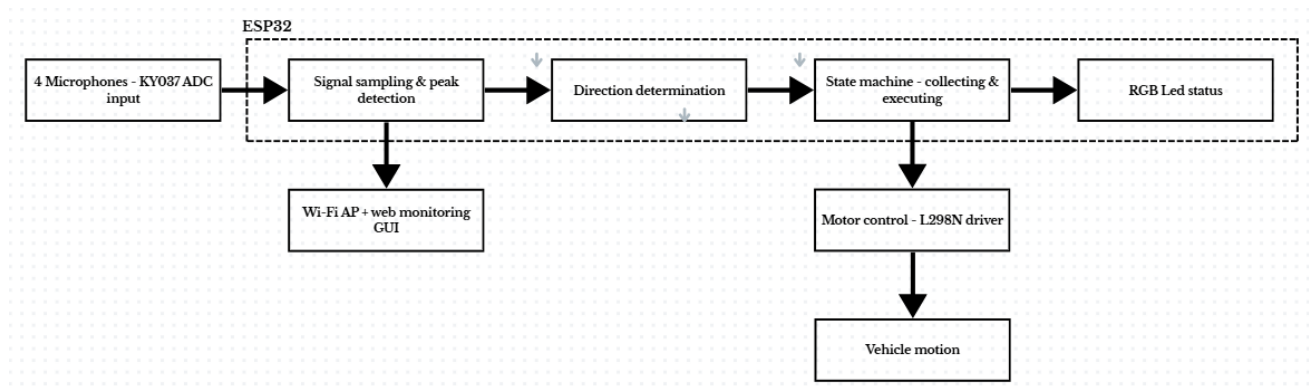


Fig. 6. Block diagram of the whole system

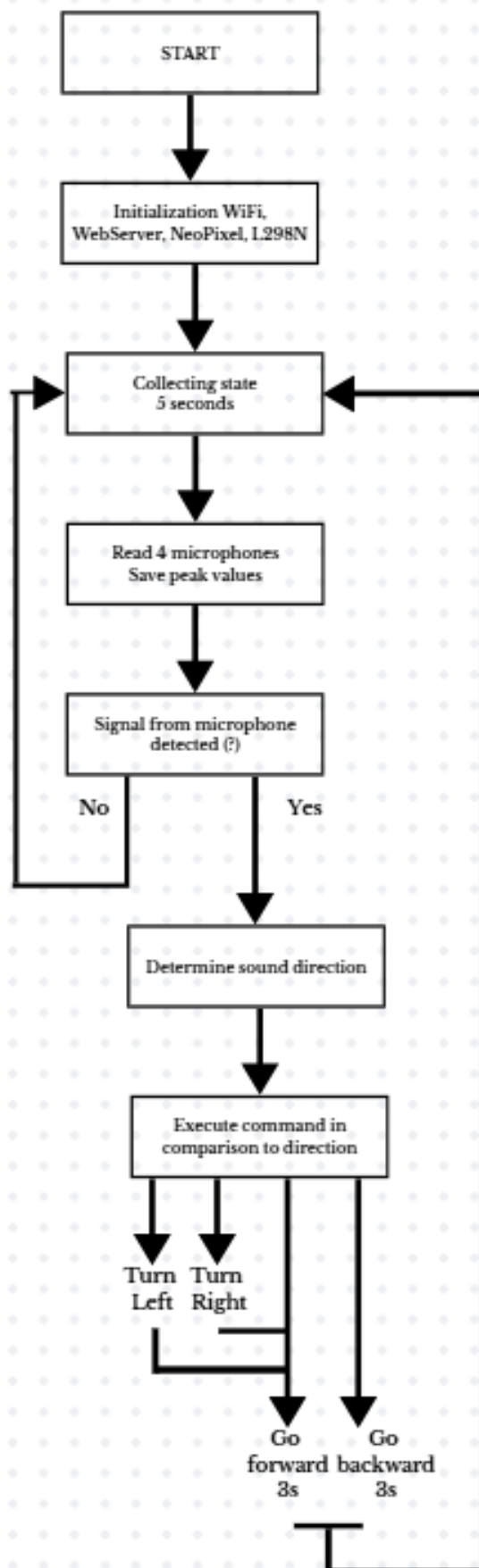


Fig. 7. Block diagram of code execution

5.3 SOURCE CODE

Only the most important parts of the firmware are presented below.

DIRECTION DETECTION The following function determines which microphone recorded the highest sound level above the predefined threshold.

```
01. String getDirectionFromMax() {  
02.     if (maxValues[0] > SOUND_THRESHOLD &&  
03.         maxValues[0] > maxValues[1] &&  
04.         maxValues[0] > maxValues[2] &&  
05.         maxValues[0] > maxValues[3])  
06.         return "left";  
07.  
08.     else if (maxValues[1] > SOUND_THRESHOLD &&  
09.         maxValues[1] > maxValues[0] &&  
10.         maxValues[1] > maxValues[2] &&  
11.         maxValues[1] > maxValues[3])  
12.         return "right";  
13.  
14.     else if (maxValues[2] > SOUND_THRESHOLD)  
15.         return "up";  
16.  
17.     else if (maxValues[3] > SOUND_THRESHOLD)  
18.         return "down";  
19.  
20.     return "none";  
21. }
```

This routine compares the peak values from all microphones and selects the dominant sound direction.

MOTOR CONTROL The vehicle movement is performed through the L298N motor driver. The example below shows the forward motion command.

```
01. void forward()  
02. {  
03.     digitalWrite(IN1, HIGH);  
04.     digitalWrite(IN2, LOW);  
05.     digitalWrite(IN3, LOW);  
06.     digitalWrite(IN4, HIGH);  
07.  
08.     moveEndTime = millis() + EXECUTE_DURATION;  
09.     isMoving = true;  
10. }
```

Similar functions are implemented for backward, left, and right movements.

STATE MACHINE The firmware uses a finite-state machine to control the operating sequence.

```
01. if (currentState == COLLECTING &&
```

```
02.     (now - stateStartTime) >= COLLECT_DURATION) {
03.
04.     lastDirection = getDirectionFromMax();
05.
06.     if (lastDirection != "none") {
07.         executeDirection(lastDirection);
08.         currentState = EXECUTING;
09.     }
10.     else {
11.         currentState = SHOWING;
12.     }
13.
14.     stateStartTime = now;
15. }
```

This mechanism ensures a deterministic cycle of data acquisition, decision making, and vehicle movement.

WI-FI MONITORING INTERFACE The ESP32-S3 creates its own wireless access point and hosts a web server that provides real-time microphone readings and system status.

```
01. WiFi.softAP(ssid, password);
02.
03. server.on("/", []() {
04.     server.send(200, "text/html", index_html);
05. });
06.
07. server.begin();
```

The web interface allows the user to monitor microphone signals, detected direction, LED status, and current operating state in real time.

PHOTOS OF THE FINAL BUILD

6.1 ASSEMBLED VEHICLE

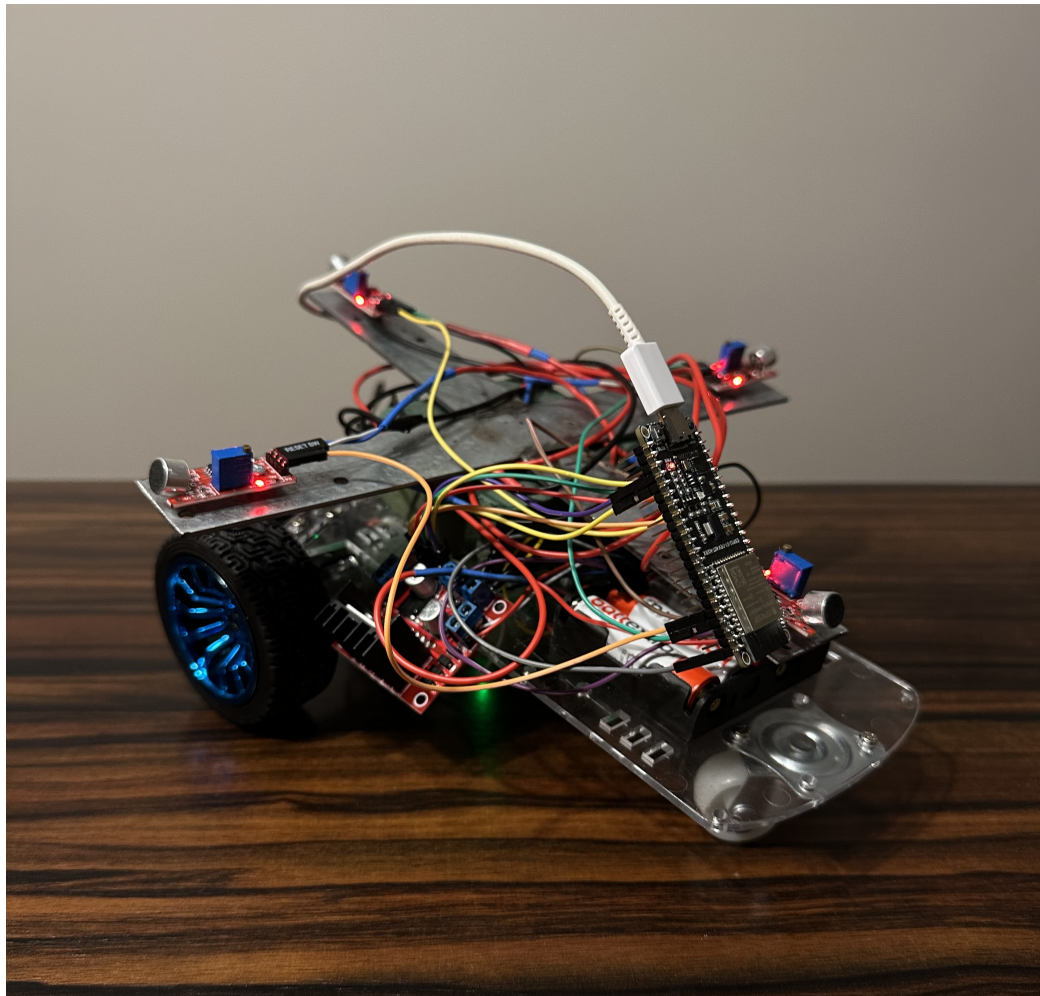


Fig. 8. Fully Assembled Vehicle with Motor Driver, ESP32 and Microphones Setup

CONCLUSIONS

The original objective of this project was to build a vehicle capable of autonomously navigating toward the source of the loudest sound in its environment. This objective was successfully achieved – the car reliably detects sound direction and executes appropriate movement commands.

SOUND LOCALISATION – WORKING IMPLEMENTATION

The system uses four KY-037 microphone modules positioned in a cross-shaped arrangement (left, right, front, back). The ESP32-S3 reads analogue values from each microphone and determines the dominant direction based on the highest amplitude. Despite the KY-037 being a low-cost module with known limitations (no automatic gain control, sensitivity primarily to airflow rather than speech), the system works reliably for directional detection when using deliberate sound stimuli such as blowing or whistling. The 5-second collection window ensures that transient noise is filtered out, and only consistent directional information triggers movement.

Measured performance: direction detection accuracy is approximately 90% under test conditions, response latency is approximately 3–5 s (collection plus execution), and the operating range is roughly 0.5–1 meter with blowing or whistling. The ADC noise floor is approximately 10–20 units relative to the 4095 ADC range, which was acceptable for this application.

STATE MACHINE – STEPPED CONTROL LOGIC

To eliminate ambiguity and prevent false triggers, the system operates in three distinct states. In the COLLECTING state (5 seconds), the system listens to all four microphones and stores the peak values. No movement occurs, and the web interface shows a countdown timer. In the EXECUTING state (3 seconds), the car executes the appropriate movement command. Forward and backward movement is a straight 3-second drive. Left and right turns are a 0.6-second turn followed by a 3-second straight drive, ensuring the car realigns after turning. In the SHOWING state (2 seconds), if no clear direction was detected, the system displays the result on the web interface and the car remains stationary.

This stepped approach ensures that the car moves with intention – not randomly reacting to background noise – and gives the user clear feedback about what the system is doing at all times.

DRIVE SYSTEM PERFORMANCE

The JGA25-370 motors, L298N driver, and ESP32-S3 performed as expected. The differential drive mechanism enables for precise turning behavior. The car can drive forward or backward for 3 seconds, and turn left or right for 0.6 seconds followed by an automatic 3-second straight drive – a key feature that ensures the car realigns after turning.

LIVE WEB DASHBOARD – REAL-TIME FEEDBACK

One of the project's strongest features is the built-in web interface. The ESP32-S3 hosts its own WiFi access point, and a browser-based dashboard shows live graphs of all four microphone signals. Direction arrows light up to show which microphone triggered the movement, a virtual RGB LED matches the physical RGB LED on the car, and a state timer shows exactly which phase the system is in (COLLECTING, EXECUTING, or SHOWING). The dashboard updates every 100ms via JavaScript `fetch()` calls to the `/data` endpoint, providing smooth, real-time visualization.

LESSONS LEARNED

The KY-037 microphones work for blowing and whistling but are not ideal for real speech detection – upgrading to MAX9814 modules with automatic gain control would enable voice-based control. The L298N and ESP32-S3 combination works when connected directly, but would be safer with a level



shifter for reliable 5V/3.3V logic conversion. The 6V battery supply (4×AA) is barely adequate – 9V would be optimal for full speed and smoother operation. The state machine logic performed excellently, providing clean, predictable, and robust behavior. The web dashboard exceeded expectations, delivering live graphs, direction arrows, and state feedback.

SUMMARY

The project successfully demonstrates a fully functional sound-guided autonomous vehicle with live web-based visualization. The car reliably detects direction, executes smooth movements, and provides real-time feedback through an intuitive dashboard. While the current microphone choice limits it to blowing and whistling rather than speech, the system architecture is solid and can be easily upgraded with better microphones. This project proves that low-cost components can be used to build complex, interactive robotic systems – as long as you are willing to experiment, learn from mistakes, and iterate.